

ZERO-TO-JOB-READY DATA ANALYST 2026

M05 - POWER BI MODELING

Power Query -> grain -> star schema -> relationships -> trust -> refresh

PROFILE -> PREPARE -> MODEL -> FILTER-TEST -> RECONCILE -> REFRESH

A polished visual cannot repair a wrong grain or relationship.

This book is your teacher

- You already know Excel and SQL, but Power BI Desktop is a new tool. Every new interface area and modeling word is introduced before use.
- Use Transform Data so you inspect/profile before loading blindly. Preserve source workbooks; transformations belong in Power Query.
- Write table grain and prove one-side key uniqueness before creating relationships. Do not use many-to-many or bidirectional filtering simply to silence a problem.
- Videos/references show the interface; this book controls the exact course data, quality traps, validation checks and recovery steps.
- Static artifacts can be verified here, but actual relationship/filter/refresh behavior must be executed in Power BI Desktop before release.

Normal PBM lesson loop

1. Read mental model + glossary
2. Open exact source workbook
3. Watch/read mapped source if useful
4. Follow Power BI steps once
5. Save screenshot/row/key evidence
6. Close help + changed retry
7. Reconcile + filter-test
8. Explain and mark only with evidence

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Module prerequisite and tool map

0. Prerequisite

M00-M04 established grain, data quality, Excel cleaning/Power Query exposure, and SQL key/cardinality/reconciliation thinking. Power BI modeling itself is not assumed. DAX/reporting is intentionally deferred to M06.

Tools / files

- Power BI Desktop on Windows - use the current official Microsoft installer/download page.
- C1_M05_POWERBI_MODELING_PRACTICE_DATA_RC1.xlsx - source data for PBM0-PBM7.
- C1_M05_POWERBI_MODELING_PRACTICE_GUIDE_RC1.pdf - exact follow-along/changed practice route.
- Protected practice review - post-attempt only.
- Gate A/B/C data workbooks + Gate guide/review PDFs - conditional state machine.
- Your PBIX files are learner-created evidence and are not pre-solved inside this package.

M05 boundary

- Modeling first; no advanced DAX/report-design dependency.
- Temporary visuals are used only to prove filter/model behavior.
- Do not hide unresolved bad keys/dates/discounts just to make a visual clean.
- Power BI Desktop live execution remains required before official learner release.

Exact practice, review and Gate route

- PBM0 import/orient -> PBM1 profile -> PBM2 transform/type -> PBM3 fact/dimension/star -> PBM4 relationships/filter flow -> PBM5 Date -> PBM6 trust checks -> PBM7 refresh/hygiene.
- For each lesson, use the exact source workbook, save PBIX/version + evidence, then attempt the changed task before protected review.
- Protected review opens only after a genuine saved attempt. Close it after repair and complete a changed retry.
- After PBM0-PBM7 mastery, attempt Gate A on RepairHub Jobs. A clean pass ends M05.
- Gate B LearnNow Subscriptions exists only after failed A + Review A + repair + explicit assignment. Gate C FreshCart Deliveries is the same conditional pattern after failed B.

Critical failures

Wrong grain; duplicate/blank one-side key; arbitrary many-to-many/bidirectional repair; hidden unmatched/invalid values; Date coverage failure; un-reconciled row/total; or failed filter/refresh proof can block a pass even when a PBIX looks polished.

PBM0 - Install, orient and import

Practice: matching PBM block in C1_M05_POWERBI_MODELING_PRACTICE_GUIDE_RC1.pdf

1. Why this lesson matters

A Power BI report can look correct while its preparation/model is wrong. This lesson builds one trust layer so later DAX/reporting is based on data whose grain, relationships and refresh behavior can be defended.

2. What you are learning

Install/orient Power BI Desktop, import the supplied workbook through Transform Data, identify the five model tables and save a PBIX.

3. What you already need / prerequisite

No prior Power BI Desktop skill is assumed.

4. Picture it in your head

Source workbook -> Navigator -> Transform Data -> Power Query Editor -> Close & Apply -> Model view -> temporary validation visual.

5. New words before we use them

Power BI Desktop = Windows authoring application. PBIX = Power BI project/report file. Report view = visuals. Data view = loaded table inspection. Model view = relationships. Power Query Editor = repeatable data preparation before load.

6. Watch / visual source

RECOMMENDED_VISUAL. Chandoo - Data Modelling & Star Schema for Power BI | 09:40-11:49. Why use it: See Get Data/import/model orientation before reproducing the course import path. Internal fallback: PBM0 learner book + practice guide fully explain views, Transform Data and PBIX save evidence.

PBM0 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Power BI Desktop has four working areas you must distinguish: Report view for visuals, Data view for loaded data inspection, Model view for tables/relationships, and Power Query Editor for repeatable preparation before load. A PBIX is the model/report project file. Use Get Data -> Excel -> Transform Data so you inspect the source before loading it blindly. Teacher action: Use the practice workbook and complete P-PBM0: Import a changed workbook containing Orders, Products, Stores and Calendar.

8. What you should see / be able to say

PBIX saved; all intended tables imported through Transform Data; learner can name the four working areas correctly.

9. Why it worked

The action preserves source evidence, makes grain/model assumptions explicit and creates a testable checkpoint. Power BI output is trusted only when independent row/key/total/filter/refresh controls agree.

10. Common beginner mistakes

Loading sheets directly without inspecting; confusing Report/Data/Model/Power Query views; trusting auto-created relationships; forgetting to save a PBIX/version before experimenting.

11. If you get stuck - recovery ladder

If the wrong sheet/query was loaded, remove the unintended query and reconnect through Transform Data. If Power BI auto-created relationships, do not trust them yet; PBM4 will rebuild relationships deliberately. 1) Restate grain/goal. 2) Identify the exact Power BI area or query. 3) Reveal only the next structural action. 4) Open matching protected review only after a saved attempt. 5) Close review and solve a changed retry.

PBM0 - Do alone, validate and master

12. Now you do a small version alone

Import a changed workbook containing Orders, Products, Stores and Calendar. Identify which view/editor you would use for each of: cleaning, relationship inspection, and a temporary model check.

13. Validate your result

PBIX saved; all intended tables imported through Transform Data; learner can name the four working areas correctly.

14. Teach it back in your own words

Without the lesson open, explain install, orient and import: the business purpose, one common failure, one validation check and how you would recover.

15. Protected review + changed fresh retry

Save the PBM0 PBIX/evidence first. If repair is needed, open the protected PBM0 review, compare only the weak competency, close it, and complete a changed source/model task. Append the retry instead of overwriting the earlier evidence.

16. Evidence / mastery + exact next route

Save PBIX/version name, model/query screenshot(s), grain/relationship notes, validation evidence, unresolved-quality log, confidence 1-5 and a short own-words explanation. PASS with evidence -> continue. PARTIAL -> targeted repair + fresh changed retry. FAIL -> return only to the named weak competency; do not restart the module.

PBM1 - Power Query interface and profiling

Practice: matching PBM block in C1_M05_POWERBI_MODELING_PRACTICE_GUIDE_RC1.pdf

1. Why this lesson matters

A Power BI report can look correct while its preparation/model is wrong. This lesson builds one trust layer so later DAX/reporting is based on data whose grain, relationships and refresh behavior can be defended.

2. What you are learning

Produce a profiling note for every source table before changing it, including grain, row count, key/blank/type/business-value issues.

3. What you already need / prerequisite

Prerequisite: PBM0. The new interface/model concept is still explained from first principles.

4. Picture it in your head

For each query: Grain -> Rows -> Required blanks -> Key uniqueness -> Types -> Business validity -> Consistency -> only then transformations.

5. New words before we use them

Profiling = inspect data before changing it. Column quality = valid/error/empty overview. Distribution = distinct/unique pattern. Column profile = detailed statistics/value distribution. Applied Steps = ordered transformation pipeline.

6. Watch / visual source

RECOMMENDED_REFERENCE. Microsoft Learn - Using the data profiling tools | Column quality, distribution and profile. Why use it: Current official explanation of Power Query profiling controls; use the visual companion to see the editor in motion. Optional visual: Chandoo - End to End Power Query Data Cleaning Masterclass | 11:40-23:59. Internal fallback: PBM1 learner book + practice guide define the exact profiling checklist.

PBM1 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Profile before transforming. For every table, state its intended grain, row count, required-key completeness, one-side key uniqueness, type problems, suspicious business values and formatting inconsistencies. Applied Steps are an ordered preparation pipeline; each change should be explainable and reversible by editing the step rather than manually changing the raw workbook. Teacher action: Use the practice workbook and complete P-PBM1: Profile a fresh Customers query: state grain, count blank IDs, test uniqueness before/after trim, identify one suspicious type/value and write the next safe action for each issue.

8. What you should see / be able to say

Profiling note exists before transformations and distinguishes duplicate-key suspicion from proven duplicate records.

9. Why it worked

The action preserves source evidence, makes grain/model assumptions explicit and creates a testable checkpoint. Power BI output is trusted only when independent row/key/total/filter/refresh controls agree.

10. Common beginner mistakes

Transforming before writing grain/row counts; profiling only the first 1,000 rows without noticing the setting; deleting a suspected duplicate before proving it; confusing type validity with business validity.

11. If you get stuck - recovery ladder

If profiling feels vague, start with grain + row count + required keys. For a suspected duplicate, compare the full business row and normalized key before deleting anything. 1) Restate grain/goal. 2) Identify the exact Power BI area or query. 3) Reveal only the next structural action. 4) Open matching protected review only after a saved attempt. 5) Close review and solve a changed retry.

PBM1 - Do alone, validate and master

12. Now you do a small version alone

Profile a fresh Customers query: state grain, count blank IDs, test uniqueness before/after trim, identify one suspicious type/value and write the next safe action for each issue.

13. Validate your result

Profiling note exists before transformations and distinguishes duplicate-key suspicion from proven duplicate records.

14. Teach it back in your own words

Without the lesson open, explain power query interface and profiling: the business purpose, one common failure, one validation check and how you would recover.

15. Protected review + changed fresh retry

Save the PBM1 PBIX/evidence first. If repair is needed, open the protected PBM1 review, compare only the weak competency, close it, and complete a changed source/model task. Append the retry instead of overwriting the earlier evidence.

16. Evidence / mastery + exact next route

Save PBIX/version name, model/query screenshot(s), grain/relationship notes, validation evidence, unresolved-quality log, confidence 1-5 and a short own-words explanation. PASS with evidence -> continue. PARTIAL -> targeted repair + fresh changed retry. FAIL -> return only to the named weak competency; do not restart the module.

PBM2 - Transform safely and set data types

Practice: matching PBM block in C1_M05_POWERBI_MODELING_PRACTICE_GUIDE_RC1.pdf

1. Why this lesson matters

A Power BI report can look correct while its preparation/model is wrong. This lesson builds one trust layer so later DAX/reporting is based on data whose grain, relationships and refresh behavior can be defended.

2. What you are learning

Prepare the fact/dimensions with justified Power Query steps while preserving raw data and documenting every row-count change.

3. What you already need / prerequisite

Prerequisite: PBM1. The new interface/model concept is still explained from first principles.

4. Picture it in your head

Raw value -> justified transformation -> explicit data type -> quality check -> prepared value. Every row-count change gets a reason.

5. New words before we use them

Data type = how a value is interpreted, such as text/date/whole/decimal. Exact duplicate = identical row at the stated grain. Business validity = value obeys business rule; type validity alone does not prove it. Applied Step = repeatable transformation.

6. Watch / visual source

RECOMMENDED_REFERENCE. Microsoft Learn - Data Types in Power Query | Detection, definition and conversion. Why use it: Current official type behavior before the course's safe-transformation lab. Optional visual: Chandoo - End to End Power Query Data Cleaning Masterclass | 24:00-40:59. Internal fallback: PBM2 learner book + guide teach the exact duplicate/date/discount/unmatched-key workflow.

PBM2 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

A safe transformation is repeatable, justified and testable. Trim formatting noise, standardize only confirmed equivalent values, set business-appropriate types, remove an exact duplicate only after proving it is exact at the stated grain, and keep unresolved unmatched/invalid values visible. Type validity and business validity are different: 1.20 is numeric but may violate a 0%-100% discount rule. Teacher action: Use the practice workbook and complete P-PBM2: Clean a fresh fact query containing one exact duplicate, one invalid date, one 130% discount and one unknown product key.

8. What you should see / be able to say

Every row-count change is explained; invalid/unmatched values remain visible; no raw-sheet manual edit is required.

9. Why it worked

The action preserves source evidence, makes grain/model assumptions explicit and creates a testable checkpoint. Power BI output is trusted only when independent row/key/total/filter/refresh controls agree.

10. Common beginner mistakes

Replacing all errors with zero/blank; deleting rows without a reason; changing raw Excel source manually; removing a duplicate before proving exact equality at the intended grain; hiding unknown keys.

11. If you get stuck - recovery ladder

If a type change creates errors, inspect the original values and isolate the failure instead of replacing all errors with zero/blank. If row count changed unexpectedly, step backward through Applied Steps until the count changes. 1) Restate grain/goal. 2) Identify the exact Power BI area or query. 3) Reveal only the next structural action. 4) Open matching protected review only after a saved attempt. 5) Close review and solve a changed retry.

PBM2 - Do alone, validate and master

12. Now you do a small version alone

Clean a fresh fact query containing one exact duplicate, one invalid date, one 130% discount and one unknown product key. Keep a before/after row-count note and a quality log.

13. Validate your result

Every row-count change is explained; invalid/unmatched values remain visible; no raw-sheet manual edit is required.

14. Teach it back in your own words

Without the lesson open, explain transform safely and set data types: the business purpose, one common failure, one validation check and how you would recover.

15. Protected review + changed fresh retry

Save the PBM2 PBIX/evidence first. If repair is needed, open the protected PBM2 review, compare only the weak competency, close it, and complete a changed source/model task. Append the retry instead of overwriting the earlier evidence.

16. Evidence / mastery + exact next route

Save PBIX/version name, model/query screenshot(s), grain/relationship notes, validation evidence, unresolved-quality log, confidence 1-5 and a short own-words explanation. PASS with evidence -> continue. PARTIAL -> targeted repair + fresh changed retry. FAIL -> return only to the named weak competency; do not restart the module.

PBM3 - Fact tables, dimensions, grain and star schema

Practice: matching PBM block in C1_M05_POWERBI_MODELING_PRACTICE_GUIDE_RC1.pdf

1. Why this lesson matters

A Power BI report can look correct while its preparation/model is wrong. This lesson builds one trust layer so later DAX/reporting is based on data whose grain, relationships and refresh behavior can be defended.

2. What you are learning

Classify fact/dimensions, state grain, make intended one-side keys unique and arrange a clear star schema.

3. What you already need / prerequisite

Prerequisite: PBM2. The new interface/model concept is still explained from first principles.

4. Picture it in your head

Fact (many event rows) <- foreign keys -> Dimensions (unique descriptive keys). Grain is written before relationship creation.

5. New words before we use them

Fact = events/measurements at a consistent grain. Dimension = descriptive entity used to filter/group. Grain = what one row represents. Star schema = fact in center with dimensions around it. One-side key = unique dimension identifier.

6. Watch / visual source

RECOMMENDED_VISUAL. Chandoo - Data Modelling & Star Schema for Power BI | 00:43-09:39. Why use it: See fact/dimension/star-schema thinking visually before classifying the course tables. Internal fallback: PBM3 learner book + guide teach grain, fact/dimension roles and one-side uniqueness.

[Open primary/reference](#)

PBM3 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

A fact table records events/measurements at a consistent grain and usually has many rows. Dimensions describe entities used to group/filter and require a unique key on the one side. A star schema keeps the fact in the center with Product/Customer/Region/Date-like dimensions around it. Model clarity begins with one sentence describing the grain of every table. Teacher action: Use the practice workbook and complete P-PBM3: Given Orders, Products, Stores and Calendar, state the grain of each, choose fact/dimensions and prove the intended one-side keys are unique before drawing the star.

8. What you should see / be able to say

Every table has a written grain; intended one-side keys are unique; fact is central in a recognizable star.

9. Why it worked

The action preserves source evidence, makes grain/model assumptions explicit and creates a testable checkpoint. Power BI output is trusted only when independent row/key/total/filter/refresh controls agree.

10. Common beginner mistakes

Calling every lookup-like table a dimension without checking key uniqueness; building relationships before grain statements; forcing many-to-many because the one side is dirty; mixing fact/dimension responsibilities.

11. If you get stuck - recovery ladder

If the dimension key is not unique, do not create many-to-many as a shortcut. Identify whether whitespace, duplicate business entities or wrong grain caused the duplicate. 1) Restate grain/goal. 2) Identify the exact Power BI area or query. 3) Reveal only the next structural action. 4) Open matching protected review only after a saved attempt. 5) Close review and solve a changed retry.

PBM3 - Do alone, validate and master

12. Now you do a small version alone

Given Orders, Products, Stores and Calendar, state the grain of each, choose fact/dimensions and prove the intended one-side keys are unique before drawing the star.

13. Validate your result

Every table has a written grain; intended one-side keys are unique; fact is central in a recognizable star.

14. Teach it back in your own words

Without the lesson open, explain fact tables, dimensions, grain and star schema: the business purpose, one common failure, one validation check and how you would recover.

15. Protected review + changed fresh retry

Save the PBM3 PBIX/evidence first. If repair is needed, open the protected PBM3 review, compare only the weak competency, close it, and complete a changed source/model task. Append the retry instead of overwriting the earlier evidence.

16. Evidence / mastery + exact next route

Save PBIX/version name, model/query screenshot(s), grain/relationship notes, validation evidence, unresolved-quality log, confidence 1-5 and a short own-words explanation. PASS with evidence -> continue. PARTIAL -> targeted repair + fresh changed retry. FAIL -> return only to the named weak competency; do not restart the module.

PBM4 - Relationships, cardinality and filter direction

Practice: matching PBM block in C1_M05_POWERBI_MODELING_PRACTICE_GUIDE_RC1.pdf

1. Why this lesson matters

A Power BI report can look correct while its preparation/model is wrong. This lesson builds one trust layer so later DAX/reporting is based on data whose grain, relationships and refresh behavior can be defended.

2. What you are learning

Create correct 1:* single-direction relationships and prove one dimension filter changes only the intended fact rows.

3. What you already need / prerequisite

Prerequisite: PBM3. The new interface/model concept is still explained from first principles.

4. Picture it in your head

Dimension[Key] (1) -> Fact[ForeignKey] (*) with single-direction filter flow. Duplicate dimension key = stop and investigate; unmatched fact key = quality issue to quantify.

5. New words before we use them

Relationship = model connection between key columns. Cardinality = 1:* / *:1 / etc. Filter direction = where filter context propagates. Active relationship = relationship used by default. Unmatched foreign key = fact key with no dimension match.

6. Watch / visual source

RECOMMENDED_VISUAL. Chandoo - Data Modelling & Star Schema for Power BI | 11:50-18:07. Why use it: See relationship creation/filter flow before performing the course's 1:* single-direction validation. Internal fallback: PBM4 learner book + guide define relationship/cardinality/filter-direction recovery.

[Open primary/reference](#)

PBM4 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Relationships encode business logic. A one-to-many relationship requires uniqueness on the intended one side; unmatched foreign keys on the fact side are a different issue. Default to single-direction filtering from dimension to fact. If Power BI refuses 1:*, investigate duplicate/blank/type-mismatched one-side keys instead of switching to many-to-many to make the warning disappear. Teacher action: Use the practice workbook and complete P-PBM4: On a changed model, create Product/Store/Date 1:* relationships to Orders and prove single-direction filter propagation with a temporary count/sum visual.

8. What you should see / be able to say

Relationship list shows 1:* and single direction; a temporary dimension filter changes only matching fact rows.

9. Why it worked

The action preserves source evidence, makes grain/model assumptions explicit and creates a testable checkpoint. Power BI output is trusted only when independent row/key/total/filter/refresh controls agree.

10. Common beginner mistakes

Using bidirectional filtering to make a visual 'work'; ignoring duplicate/blank/type-mismatched one-side keys; confusing unmatched fact keys with duplicate dimension keys; trusting relationships without a filter test.

11. If you get stuck - recovery ladder

If a relationship cannot be 1:*, inspect one-side duplicates/blanks/types. If a filter does not propagate, verify active relationship, direction and matching key values before enabling bidirectional filtering. 1) Restate grain/goal. 2) Identify the exact Power BI area or query. 3) Reveal only the next structural action. 4) Open matching protected review only after a saved attempt. 5) Close review and solve a changed retry.

PBM4 - Do alone, validate and master

12. Now you do a small version alone

On a changed model, create Product/Store/Date 1:* relationships to Orders and prove single-direction filter propagation with a temporary count/sum visual.

13. Validate your result

Relationship list shows 1:* and single direction; a temporary dimension filter changes only matching fact rows.

14. Teach it back in your own words

Without the lesson open, explain relationships, cardinality and filter direction: the business purpose, one common failure, one validation check and how you would recover.

15. Protected review + changed fresh retry

Save the PBM4 PBIX/evidence first. If repair is needed, open the protected PBM4 review, compare only the weak competency, close it, and complete a changed source/model task. Append the retry instead of overwriting the earlier evidence.

16. Evidence / mastery + exact next route

Save PBIX/version name, model/query screenshot(s), grain/relationship notes, validation evidence, unresolved-quality log, confidence 1-5 and a short own-words explanation. PASS with evidence -> continue. PARTIAL -> targeted repair + fresh changed retry. FAIL -> return only to the named weak competency; do not restart the module.

PBM5 - Date dimension and date relationships

Practice: matching PBM block in C1_M05_POWERBI_MODELING_PRACTICE_GUIDE_RC1.pdf

1. Why this lesson matters

A Power BI report can look correct while its preparation/model is wrong. This lesson builds one trust layer so later DAX/reporting is based on data whose grain, relationships and refresh behavior can be defended.

2. What you are learning

Audit the supplied Date table, connect it correctly to the fact date and explain why Month Number/date coverage matter.

3. What you already need / prerequisite

Prerequisite: PBM4. The new interface/model concept is still explained from first principles.

4. Picture it in your head

Date[Date] (1) -> Fact[Business Date] (*). Date dimension has continuous dates and separate Month + Month Number attributes.

5. New words before we use them

Date dimension = one row per date calendar table. Date key = unique/nonblank date column. Coverage = calendar spans required fact dates. Month Number = numeric sort helper. Mark as date table = Power BI configuration used in relevant time-intelligence scenarios.

6. Watch / visual source

RECOMMENDED_REFERENCE. Microsoft Learn - Set and use date tables in Power BI Desktop | Date uniqueness/nonblank/continuity and date-table setup. Why use it: More exact current official source for the lesson than a generic relationship page. Optional visual: Chandoo - Data Modelling & Star Schema for Power BI | 11:50-18:07 relationship refresher. Internal fallback: PBM5 learner book + guide teach date coverage, Month Number and Date-to-fact relationship checks.

PBM5 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

A Date dimension gives the model one consistent calendar. It should contain one row per date, a unique/nonblank Date key, no gaps in the required range and useful attributes such as Year, Month, Month Number, Quarter and Year-Month. Connect Date[Date] 1:* to the fact date and keep Month Number so month names can sort chronologically. Teacher action: Use the practice workbook and complete P-PBM5: Audit a changed calendar that spans the fact dates.

8. What you should see / be able to say

Date is unique/nonblank/continuous for the required range; Date -> fact relationship is active and correctly directed.

9. Why it worked

The action preserves source evidence, makes grain/model assumptions explicit and creates a testable checkpoint. Power BI output is trusted only when independent row/key/total/filter/refresh controls agree.

10. Common beginner mistakes

Using Month Name without Month Number; date range does not cover fact dates; duplicate/null dates; connecting Date to the wrong fact date; beginning DAX time intelligence before the model/date relationship is stable.

11. If you get stuck - recovery ladder

If Date does not cover the fact dates or contains duplicates/gaps, stop and repair/replace the calendar range before building time analysis. Do not create DAX time intelligence in this module. 1) Restate grain/goal. 2) Identify the exact Power BI area or query. 3) Reveal only the next structural action. 4) Open matching protected review only after a saved attempt. 5) Close review and solve a changed retry.

PBM5 - Do alone, validate and master

12. Now you do a small version alone

Audit a changed calendar that spans the fact dates. Prove uniqueness/nonblank/coverage, connect it to the fact date, and explain how Month Name will be sorted.

13. Validate your result

Date is unique/nonblank/continuous for the required range; Date -> fact relationship is active and correctly directed.

14. Teach it back in your own words

Without the lesson open, explain date dimension and date relationships: the business purpose, one common failure, one validation check and how you would recover.

15. Protected review + changed fresh retry

Save the PBM5 PBIX/evidence first. If repair is needed, open the protected PBM5 review, compare only the weak competency, close it, and complete a changed source/model task. Append the retry instead of overwriting the earlier evidence.

16. Evidence / mastery + exact next route

Save PBIX/version name, model/query screenshot(s), grain/relationship notes, validation evidence, unresolved-quality log, confidence 1-5 and a short own-words explanation. PASS with evidence -> continue. PARTIAL -> targeted repair + fresh changed retry. FAIL -> return only to the named weak competency; do not restart the module.

PBM6 - Model validation and trust checks

Practice: matching PBM block in C1_M05_POWERBI_MODELING_PRACTICE_GUIDE_RC1.pdf

1. Why this lesson matters

A Power BI report can look correct while its preparation/model is wrong. This lesson builds one trust layer so later DAX/reporting is based on data whose grain, relationships and refresh behavior can be defended.

2. What you are learning

Produce a validation log that proves the model did not silently lose/duplicate data and records unresolved quality issues.

3. What you already need / prerequisite

Prerequisite: PBM5. The new interface/model concept is still explained from first principles.

4. Picture it in your head

Trust = row reconciliation + key checks + unmatched-key checks + type/business checks + control total + filter-path proof.

5. New words before we use them

Reconciliation = independent agreement between source/prepared/model controls. Invariant = value/count expected not to change without reason. Filter-propagation test = prove dimension filter reaches intended fact rows. Unresolved issue = known quality risk kept visible.

6. Watch / visual source

RECOMMENDED_VISUAL. Chandoo - Data Modelling & Star Schema for Power BI | 00:43-18:07. Why use it: Revisit the complete modeling flow while building an independent trust/validation log. Internal fallback: PBM6 learner book + guide contain the seven-control validation contract.

[Open primary/reference](#)

PBM6 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

A model is trustworthy only when multiple independent checks agree. Reconcile source/prepared row counts, fact-grain uniqueness, dimension-key uniqueness, unmatched foreign keys, conversion/business-rule issues, a stable control total and a filter-propagation test. One total never proves the whole model; different checks catch different failure modes. Teacher action: Use the practice workbook and complete P-PBM6: Build a fresh model validation log with seven controls: source/prepared rows, fact-grain duplicate test, dimension uniqueness, unmatched keys, type/business issues, one monetary/count total and one filter test.

8. What you should see / be able to say

Validation log contains independent checks and reconciles the protected gross/control total without hiding quality issues.

9. Why it worked

The action preserves source evidence, makes grain/model assumptions explicit and creates a testable checkpoint. Power BI output is trusted only when independent row/key/total/filter/refresh controls agree.

10. Common beginner mistakes

Trusting one control total; ignoring key/unmatched failures because money matches; validating only after building the report; hiding unresolved issues rather than logging them.

11. If you get stuck - recovery ladder

If a control total matches but key checks fail, treat the model as untrusted. If totals differ, compare source/prepared rows and isolate the first transformation where the invariant changes. 1) Restate grain/goal. 2) Identify the exact Power BI area or query. 3) Reveal only the next structural action. 4) Open matching protected review only after a saved attempt. 5) Close review and solve a changed retry.

PBM6 - Do alone, validate and master

12. Now you do a small version alone

Build a fresh model validation log with seven controls: source/prepared rows, fact-grain duplicate test, dimension uniqueness, unmatched keys, type/business issues, one monetary/count total and one filter test.

13. Validate your result

Validation log contains independent checks and reconciles the protected gross/control total without hiding quality issues.

14. Teach it back in your own words

Without the lesson open, explain model validation and trust checks: the business purpose, one common failure, one validation check and how you would recover.

15. Protected review + changed fresh retry

Save the PBM6 PBIX/evidence first. If repair is needed, open the protected PBM6 review, compare only the weak competency, close it, and complete a changed source/model task. Append the retry instead of overwriting the earlier evidence.

16. Evidence / mastery + exact next route

Save PBIX/version name, model/query screenshot(s), grain/relationship notes, validation evidence, unresolved-quality log, confidence 1-5 and a short own-words explanation. PASS with evidence -> continue. PARTIAL -> targeted repair + fresh changed retry. FAIL -> return only to the named weak competency; do not restart the module.

PBM7 - Refresh and model hygiene

Practice: matching PBM block in C1_M05_POWERBI_MODELING_PRACTICE_GUIDE_RC1.pdf

1. Why this lesson matters

A Power BI report can look correct while its preparation/model is wrong. This lesson builds one trust layer so later DAX/reporting is based on data whose grain, relationships and refresh behavior can be defended.

2. What you are learning

Perform a controlled refresh test and prove the preparation/model remains stable without manual source edits.

3. What you already need / prerequisite

Prerequisite: PBM6. The new interface/model concept is still explained from first principles.

4. Picture it in your head

Controlled source change -> Refresh -> same preparation logic replays -> validation log reconfirms invariants -> save clean PBIX/version note.

5. New words before we use them

Refresh = rerun data retrieval/preparation/model update. Dependency = source/path/query assumption refresh relies on. Schema = column/table structure. Reproducibility = same steps work again on structurally compatible new data. Hygiene = clear names, documented assumptions and no hidden manual fix.

6. Watch / visual source

RECOMMENDED_REFERENCE. Microsoft Learn - Data refresh in Power BI | Refresh dependencies, options and best-practice concepts. Why use it: Current official refresh reference; use the visual companion for the local Power Query refresh demonstration. Optional visual: Chandoo - End to End Power Query Data Cleaning Masterclass | 1:02:00-1:07:39. Internal fallback: PBM7 learner book + guide give the controlled source-change -> refresh -> revalidation route.

PBM7 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Refresh is a reproducibility test. A good model can re-run its preparation steps against the same structure without manual repairs. Use clear query/table names, keep raw-source assumptions documented, avoid hidden local-file dependencies where possible, refresh after a controlled source change, and confirm row counts/control totals/relationships remain valid after refresh. Teacher action: Use the practice workbook and complete P-PBM7: Change one non-problem source value and add one valid fact row, refresh, then prove expected counts/totals changed while relationships and unresolved quality flags remain consistent.

8. What you should see / be able to say

After a controlled source change, refresh completes and expected row/total changes are observed with stable model logic.

9. Why it worked

The action preserves source evidence, makes grain/model assumptions explicit and creates a testable checkpoint. Power BI output is trusted only when independent row/key/total/filter/refresh controls agree.

10. Common beginner mistakes

Manually fixing loaded output after refresh; changing source structure without documenting dependency; declaring refresh success without rechecking counts/totals/relationships; confusing Service scheduled refresh with the local Desktop refresh test.

11. If you get stuck - recovery ladder

If refresh breaks, identify whether the source schema/path changed, a step depends on fragile column names/types, or a relationship lost a key. Repair the reproducibility cause rather than manually fixing output rows. 1) Restate grain/goal. 2) Identify the exact Power BI area or query. 3) Reveal only the next structural action. 4) Open matching protected review only after a saved attempt. 5) Close review and solve a changed retry.

PBM7 - Do alone, validate and master

12. Now you do a small version alone

Change one non-problem source value and add one valid fact row, refresh, then prove expected counts/totals changed while relationships and unresolved quality flags remain consistent.

13. Validate your result

After a controlled source change, refresh completes and expected row/total changes are observed with stable model logic.

14. Teach it back in your own words

Without the lesson open, explain refresh and model hygiene: the business purpose, one common failure, one validation check and how you would recover.

15. Protected review + changed fresh retry

Save the PBM7 PBIX/evidence first. If repair is needed, open the protected PBM7 review, compare only the weak competency, close it, and complete a changed source/model task. Append the retry instead of overwriting the earlier evidence.

16. Evidence / mastery + exact next route

Save PBIX/version name, model/query screenshot(s), grain/relationship notes, validation evidence, unresolved-quality log, confidence 1-5 and a short own-words explanation. PASS with evidence -> continue. PARTIAL -> targeted repair + fresh changed retry. FAIL -> return only to the named weak competency; do not restart the module.

End-of-module mastery checklist

- PBM0: Import through Transform Data, identify Report/Data/Model/Power Query areas and save PBIX evidence.
- PBM1: Profile grain, rows, keys, blanks, types, business-value issues before transformations.
- PBM2: Transform safely in Power Query, explain every row-count change and keep invalid/unmatched issues visible.
- PBM3: Classify fact/dimensions, state grain and prove intended one-side keys before star-schema arrangement.
- PBM4: Create deliberate 1:* single-direction relationships and prove filter propagation with a temporary check.
- PBM5: Audit Date uniqueness/nonblank/coverage, connect it correctly and explain chronological month sorting.
- PBM6: Build a multi-check validation log; one matching total alone does not prove model trust.
- PBM7: Change a valid source value/row, refresh, and prove preparation/model controls remain stable.

Gate readiness

Attempt Gate A only after the live PBIX can be explained and the practice controls reconcile.

A clean Gate A pass ends M05. B/C are remediation retests only.

Revision / modeling quick reference

Power Query	Inspect/profile first; transformations are repeatable Applied Steps.
Grain	Write what one row represents for every table/query.
Fact vs dimension	Facts summarize events; dimensions filter/group entities.
Star schema	Clear fact center with dimension paths; avoid accidental mixed-grain design.
1:* relationship	The 1 side must have an appropriate unique key.
Filter direction	Default dimension -> fact single direction unless a justified model requires otherwise.
Unmatched fact key	Quality issue to count; different from duplicate one-side key.
Date table	Unique/nonblank/continuous required range + sortable Month Number.
Validation	Rows + keys + unmatched + business validity + control total + filter-path proof.
Refresh	Controlled source change -> refresh -> re-run validation invariants.

Next route: Gate A -> M06 Power BI DAX / Reporting only after modeling competency pass.